

ELECTRIC GENERATION AND TRANSMISSION ON THE RAILBELT

Joe Griffith
Chief Executive Officer, ARCTEC

*Alaska Conference of Mayors Energy Summit
Juneau, Alaska
February 18, 2014*



RAILBELT MAKEUP

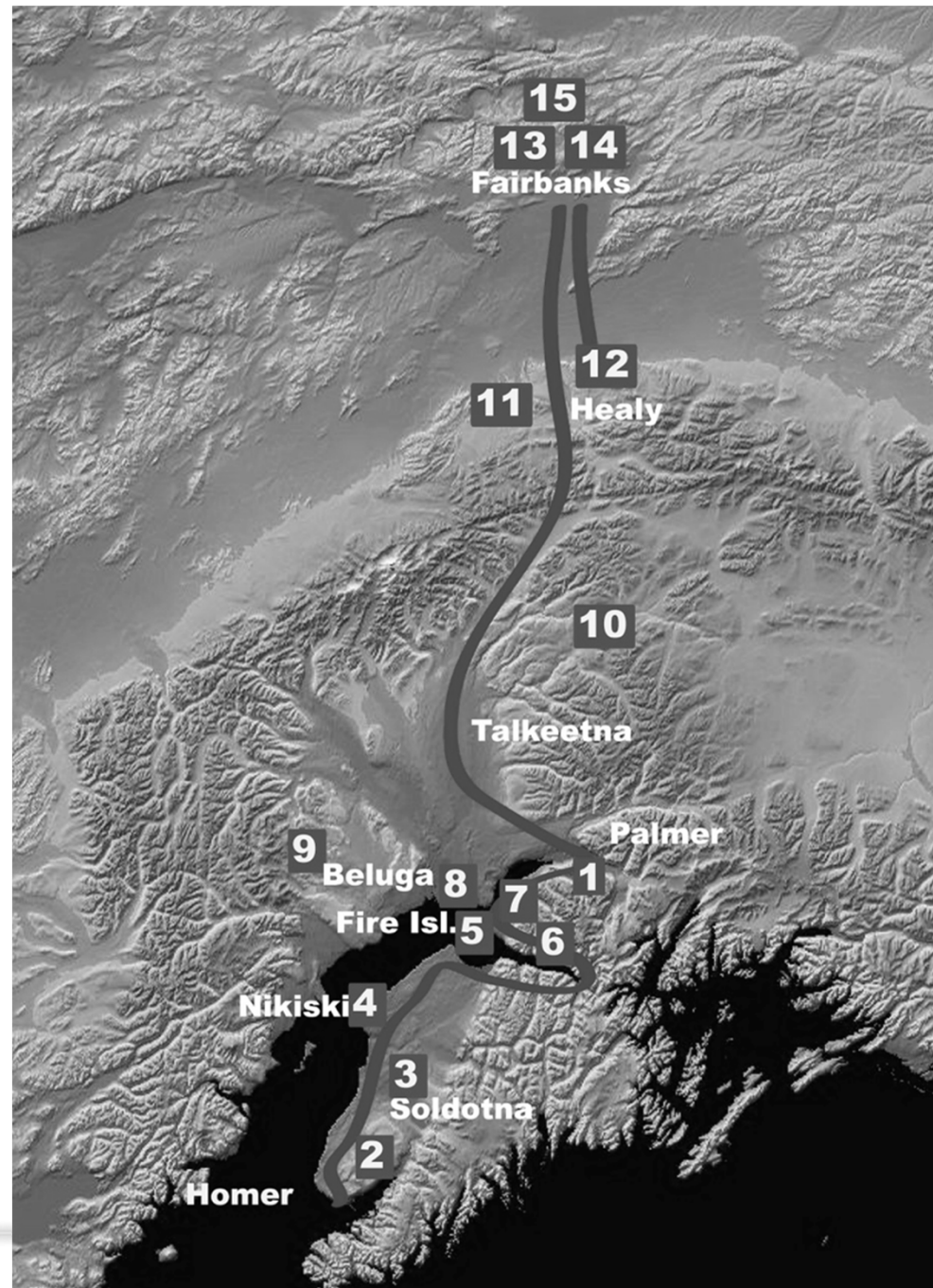
- Electrical utilities are all cooperatives or government-owned
 - Anchorage Municipal Light & Power (gov't)
 - Seward Electric System (gov't)
 - Chugach Electric Association
 - Golden Valley Electric Association
 - Homer Electric Association
 - Matanuska Electric Association

RAILBELT TRANSMISSION SYSTEM

- Connects all utilities
- Facilitates economic dispatch
- Transfers power
- Shares reserves

RAILBELT ELECTRICAL GENERATION

- 1 – Eklutna Generation Station (EGS)
- 2 – Bradley Lake
- 3 – Soldotna
- 4 – Nikiski
- 5 – Fire Island
- 6 – SPP
- 7 – ML&P Plants 2 & 2A
- 8 – Beluga
- 9 – Mount Spurr Geothermal Project
- 10 – Watana
- 11 – Eva Creek
- 12 – Healy 2 (HCC)
- 13 – Battery Energy Storage System (BESS)
- 14 – Zehnder
- 15 – North Pole



RAILBELT GENERATION SUITE

- Chugach – Beluga, South Anchorage, IGT, Cooper Lake hydro, Eklutna hydro
- AML&P – Plant 1, 2, & 2A (on drawing board), Eklutna hydro
- GVEA – North Pole, Eva Creek Wind, Healy 1 & 2, Aurora (IPP), BESS, Zehnder
- MEA – Eklutna Generation Station, Eklutna hydro, + 2 small IPPs
- HEA – Nikiski combined cycle, Soldotna, Bernice Lake
- SES – Seward plant

Note: All above participate in Bradley Lake hydro (state-owned IPP)- 124 mW capacity

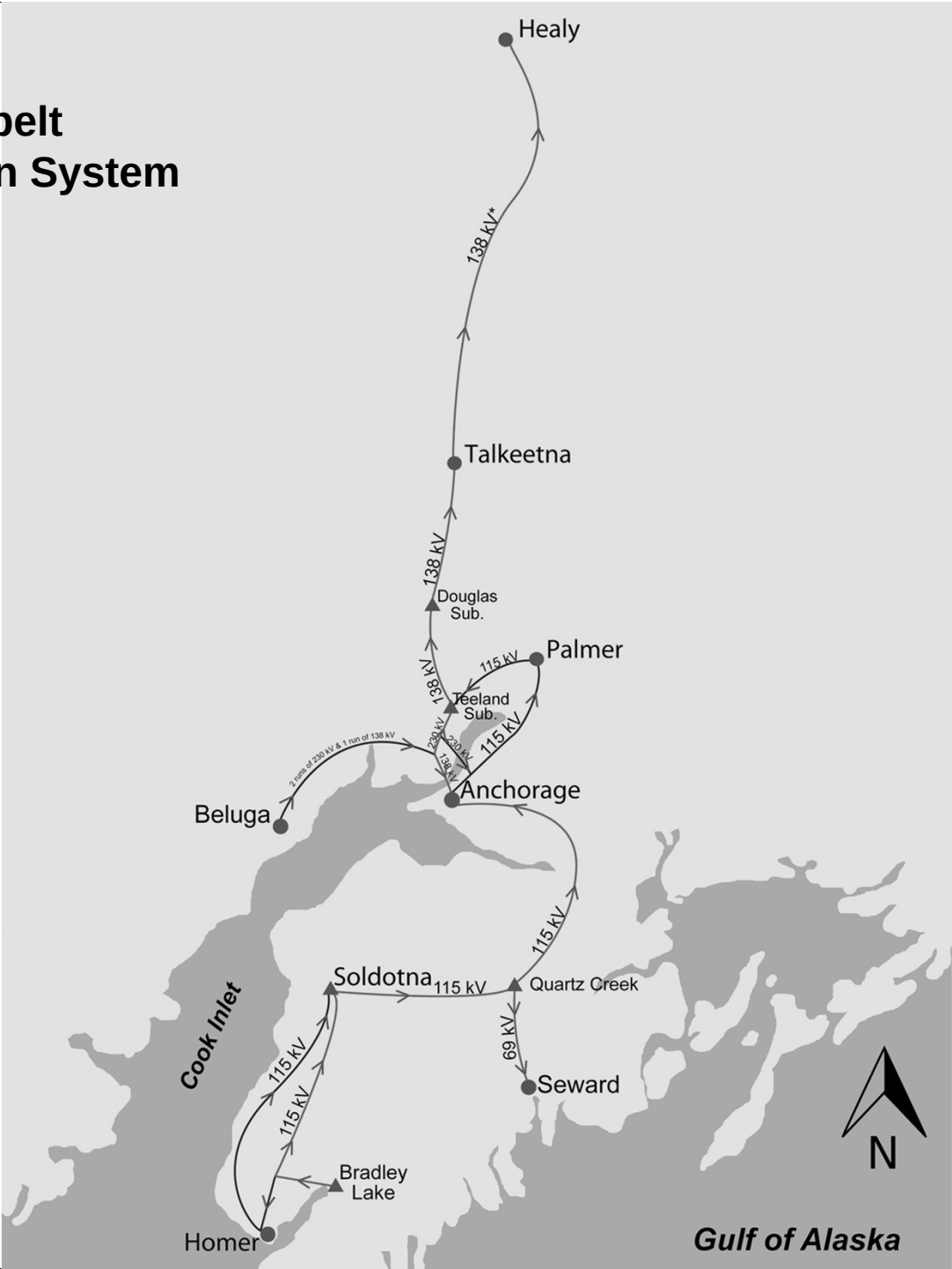
RAILBELT SITUATION

- New generation system changes playing field due to transmission constraints
 - Limits Bradley Lake power deliveries when needed most.
- Railbelt economy demands reasonably-priced power (as does the rest of Alaska).
 - RCA insists on “cost-causer, cost-payer” concept.

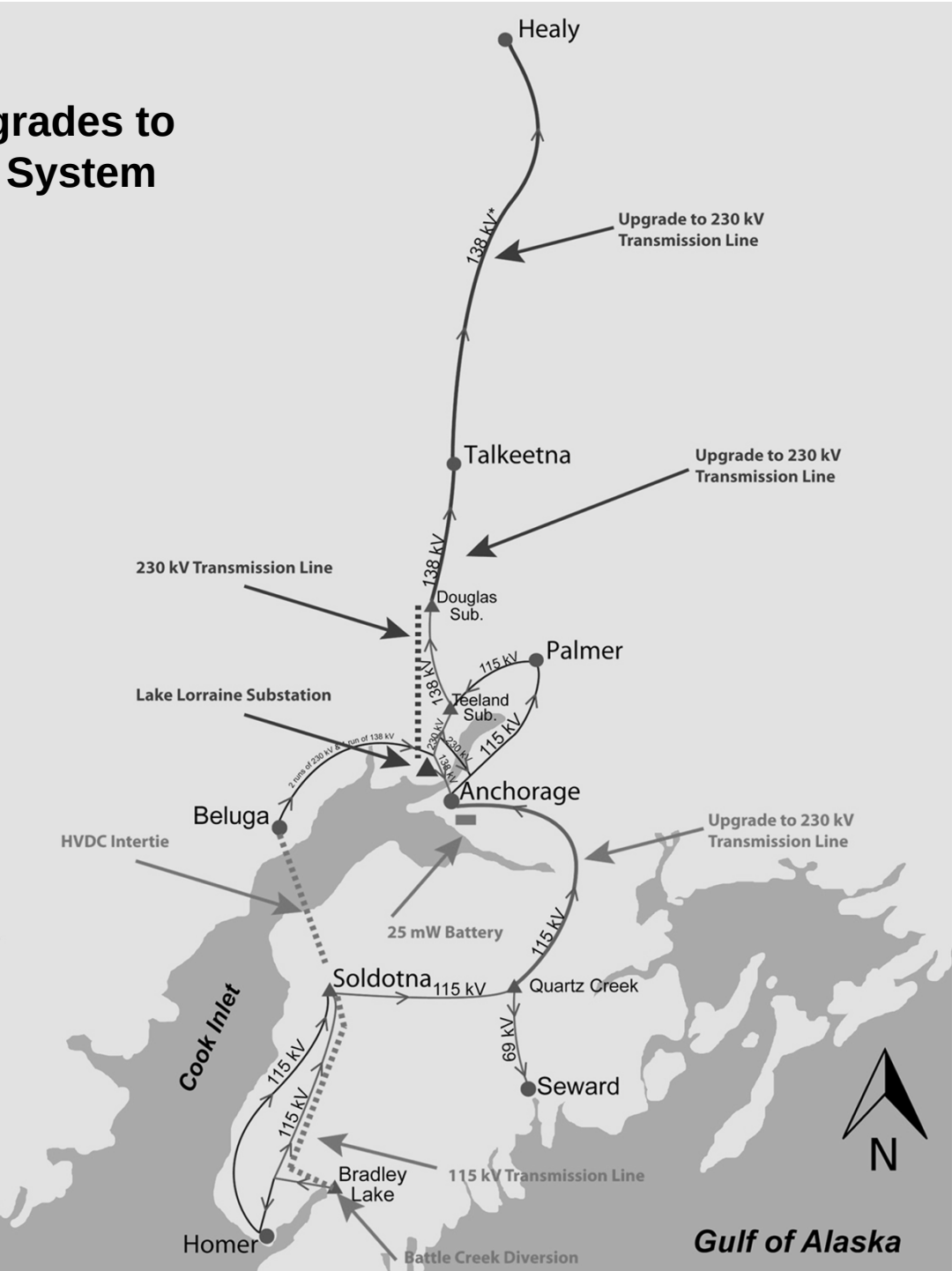
AEA TRANSMISSION STUDY (2012)

- Southern section upgrades proposed:
 - Quartz creek to University (AC)
 - Additional conductor from Quartz Creek to Daves Creek
 - 230 kV upgrade from Daves Creek to University s/s
 - DC tie line from Bernice Lake to Beluga
 - 2nd 115kV line from Bradley Junction to Soldotna
 - Anchorage battery
- Northern Upgrades
 - 230 kV Lake Lorraine to Douglas upgrade
 - Douglas to Healy upgrade to 230 kV

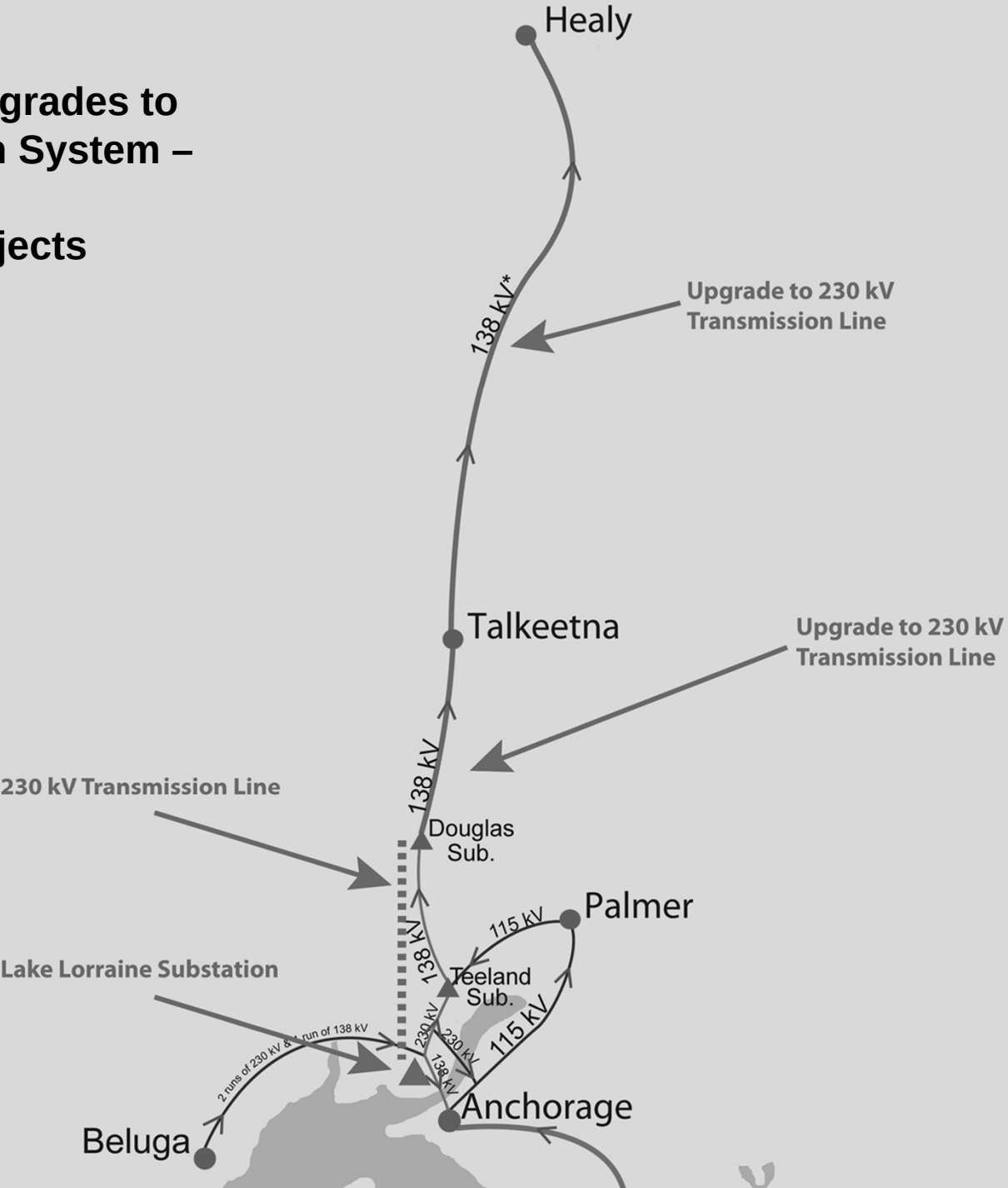
**Current Railbelt
Transmission System**



Proposed Upgrades to Transmission System

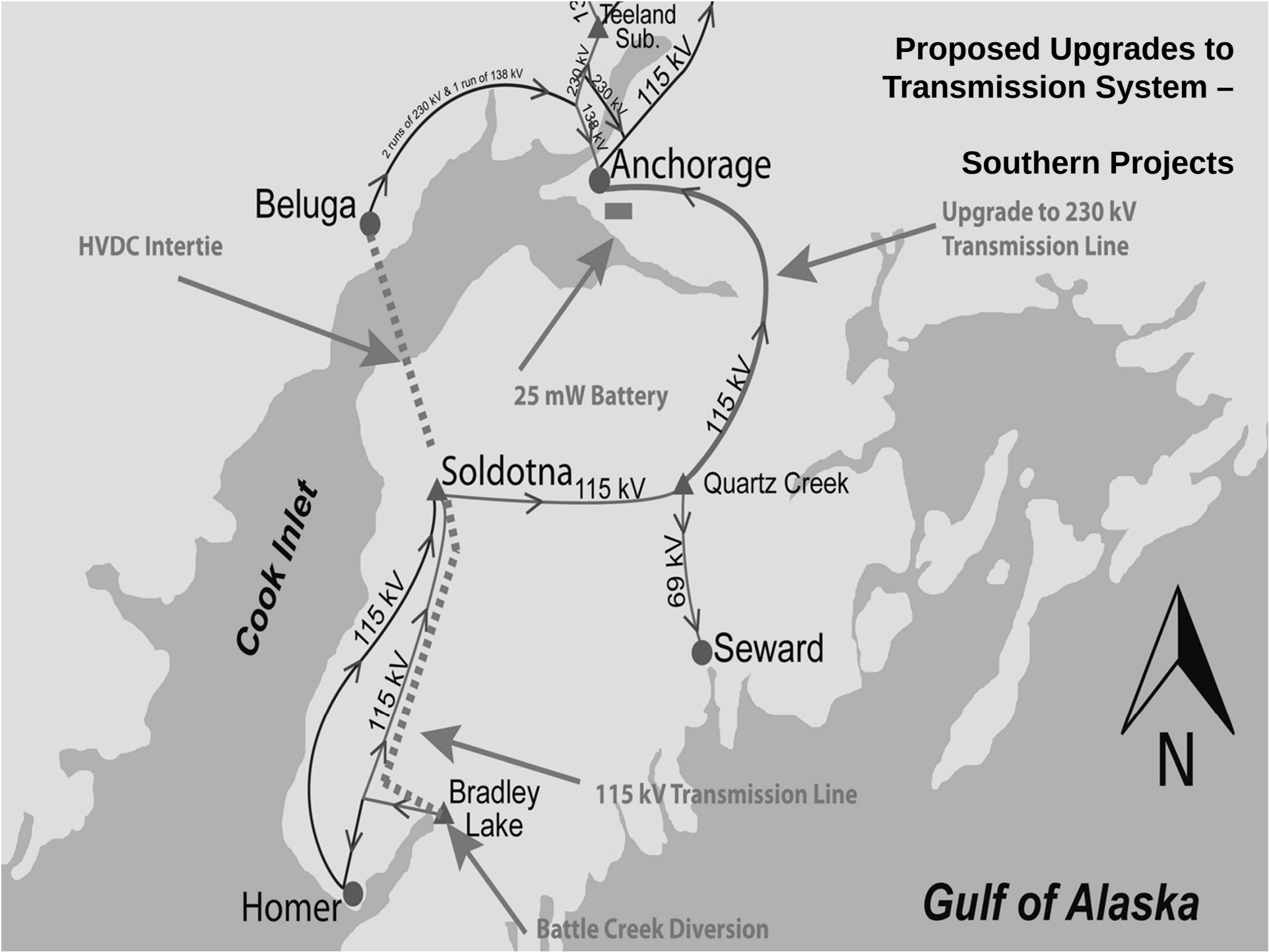


**Proposed Upgrades to
Transmission System –
Northern Projects**



Proposed Upgrades to Transmission System –

Southern Projects



TRANSCO

- A transmission-owning entity
- Regulated entity that owns, constructs, and maintains transmission lines used to transmit wholesale power
- May handle power dispatch and coordination
- Secures financing
- Executes projects

ISO

- Description
 - Owns no assets
 - Neutral party responsible for planning, management and control of the electric transmission grid in a region
- Characteristics
 - Ensures non-discriminatory access
 - Possesses operational authority for all transmission facilities under the ISO's control
 - Exclusive authority to maintain short-term reliability
 - Operates under a regulatory compact
 - Plans, conditions and approves projects

COULD BE RAILBELT MODEL

- TRANSCO = Alaska Railbelt Cooperative Transmission & Electric Company (ARCTEC)
 - Non-discriminatory open access
 - Existing assets from transmission providers are pooled for joint use with full cost recovery for all users
 - Universal rates based on system load that will encourage economic transfer of generation across network
- Transition to full implementation
 - Single control area
 - Economic dispatch
 - Centralized planning for all future transmission

CONCLUSION

- Without a program to rectify transmission shortcomings asap
 - Cost of electricity will continue to rise
 - Possible savings will be forfeited
 - Alaska will not achieve its alternative energy goal
 - Railbelt will be captive to Cook Inlet natural gas pricing and supply issues
- NEXT STEPS
 - Continue 50/50 deal between State and utilities for Bradley Lake upgrades
 - Reliability standards and *Open Access* proceeding through RCA process
 - ARCTEC proceeding with TRANSCO/ISO options

DISCUSSION / QUESTIONS